

Experiment No.	1
Experiment Title	Working with Python Packages – NumPy, Pandas, SciPy, Scikit-Learn, Matplotlib ¹
Student Name	Barath
Register Number	3122247001011
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	

1 Objective

To explore the core Python libraries used in a Machine Learning workflow – NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib – and to understand key operations such as array manipulation, data preprocessing, mathematical computing, model building, and data visualization. Further, to explore publicly available datasets from the UCI Machine Learning Repository and Kaggle, identify the type of Machine Learning task associated with each, and perform Exploratory Data Analysis (EDA) on them.

2 Problem Statement

Given five publicly available datasets – Iris, Loan Amount Prediction, Diabetes Prediction, Email Spam Classification, and MNIST Handwritten Digit Classification – the task is to load each dataset, identify the Machine Learning task it corresponds to (regression, binary classification, or multi-class classification), and perform exploratory data analysis to understand its structure, quality, and distribution. The input to the workflow is a raw dataset file (CSV); the output is a statistical and visual understanding of the dataset (distributions, missing values, correlations, class balance, outliers) that guides the choice of feature selection technique and ML algorithm for each dataset.

3 Dataset Description

Iris Dataset

Dataset Name	Iris
Dataset Source	UCI Machine Learning Repository (via <code>sklearn.datasets.load_iris</code>)
Number of Samples	150
Number of Features	4 (sepal length, sepal width, petal length, petal width)
Number of Classes	3 (setosa, versicolor, virginica – 50 each)
Missing Values	0
Train-Test Split	Not performed – exploratory analysis only

Pima Indians Diabetes Dataset

Dataset Name	Pima Indians Diabetes
--------------	-----------------------

Dataset Source	UCI Machine Learning Repository / Kaggle
Number of Samples	768
Number of Features	8 (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age)
Number of Classes	2 (Outcome: 500 non-diabetic, 268 diabetic)
Missing Values	0 null cells; however 5 Glucose and 374 Insulin values are recorded as 0, which is physiologically implausible and indicates disguised missing data
Train-Test Split	Not performed – exploratory analysis only

Loan Prediction Dataset

Dataset Name	Loan Prediction
Dataset Source	Kaggle / Analytics Vidhya
Number of Samples	614
Number of Features	11 (excluding Loan_ID and target Loan_Status)
Number of Classes	2 (Loan_Status: 422 Approved, 192 Rejected)
Missing Values	149 missing cells across 7 columns (worst: Credit_History – 50)
Train-Test Split	Not performed – exploratory analysis only

SMS Spam Collection Dataset

Dataset Name	SMS Spam Collection
Dataset Source	UCI Machine Learning Repository
Number of Samples	5,572
Number of Features	1 text feature (message content)
Number of Classes	2 (4,825 ham, 747 spam)
Missing Values	0 (403 duplicate rows present)
Train-Test Split	Not performed – exploratory analysis only

MNIST Dataset (Sample)

Dataset Name	MNIST Handwritten Digits (3,000-row sample)
Dataset Source	MNIST database (LeCun et al.), CSV mirror
Number of Samples	3,000
Number of Features	784 (28×28 grayscale pixels)
Number of Classes	10 (digits 0–9, roughly balanced – 270 to 332 each)

Missing Values	0
Train-Test Split	Not performed – exploratory analysis only

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Iris Dataset

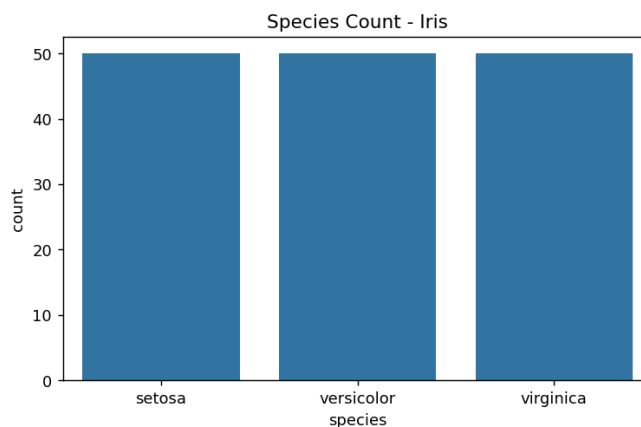


Figure 1: Species count plot – Iris

Inference: The three species (setosa, versicolor, virginica) are perfectly balanced at 50 samples each. This balance means accuracy is a reliable evaluation metric here and no class-imbalance handling (e.g. resampling) is required before training a classifier.

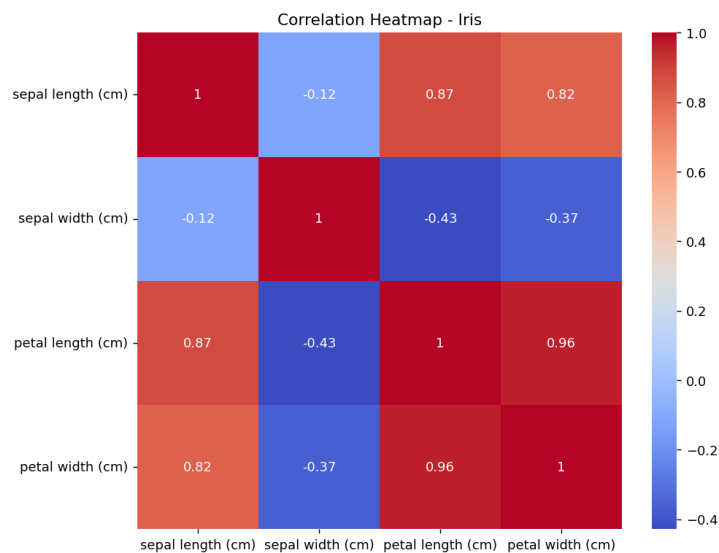


Figure 2: Correlation heatmap – Iris

Inference: Petal length and petal width are very strongly correlated with each other, and both correlate strongly with sepal length, while sepal width correlates weakly and even negatively with the other features. This indicates that petal measurements alone carry most of the discriminative signal for species classification, and the two petal features are partially redundant with one another.

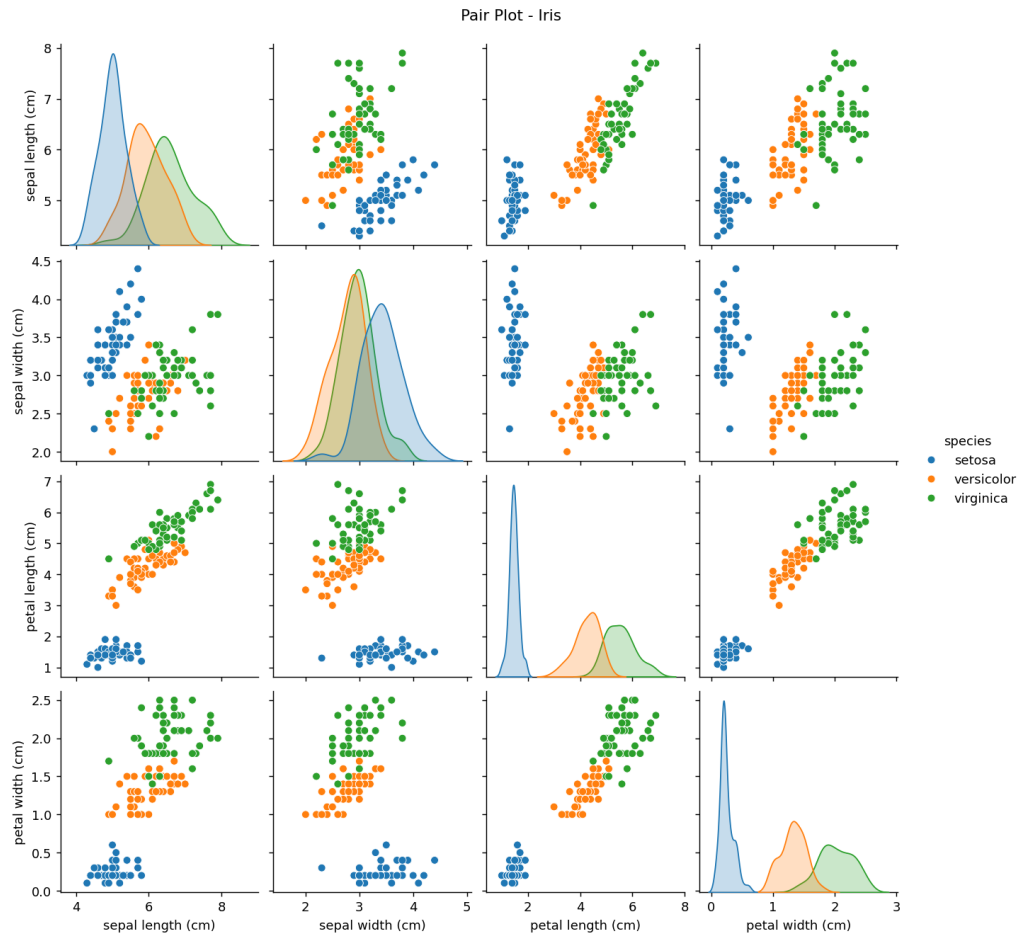


Figure 3: Pair plot coloured by species – Iris

Inference: Setosa is linearly separable from the other two species on almost every feature pair, especially petal length vs. petal width, while versicolor and virginica show some overlap. This suggests a simple linear or distance-based classifier (Logistic Regression, KNN) should already perform well, with most residual error expected between versicolor and virginica.

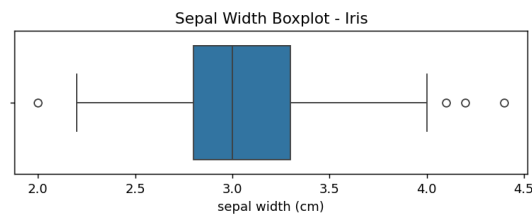


Figure 4: Sepal width box plot – Iris

Inference: The IQR method flags 4 outlier values in sepal width out of 150 samples (about 2.7%). Given the small number and that Iris is a well-curated benchmark dataset, these are most plausibly genuine measurement variation rather than data-entry errors, so retaining them is reasonable.

Diabetes Dataset

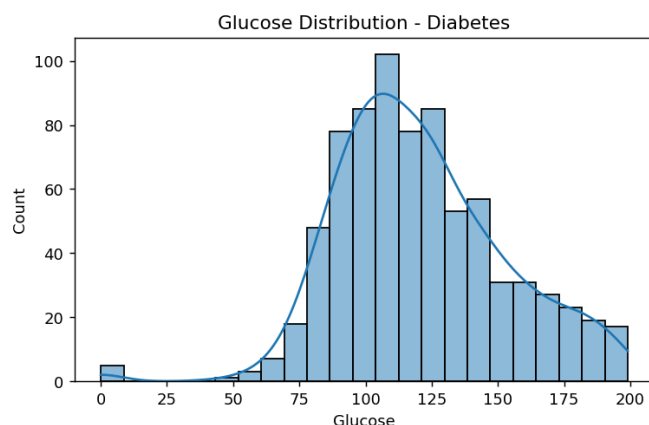


Figure 5: Glucose distribution – Diabetes

Inference: Glucose is close to a bell-shaped distribution but with a noticeable spike of zero values – 5 records have $\text{Glucose} = 0$, which is physiologically impossible and indicates the value is actually a missing-data placeholder rather than a true zero reading.

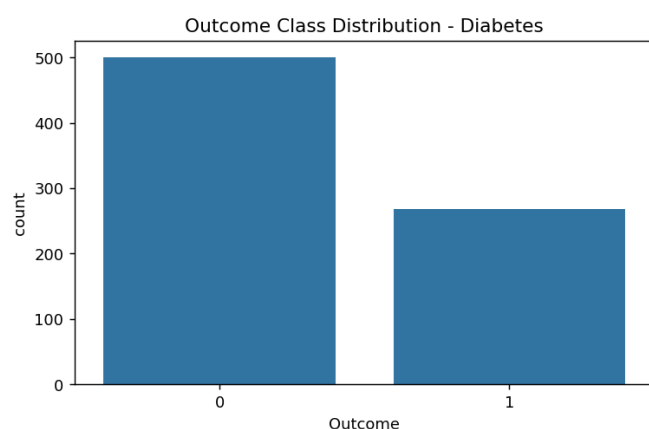


Figure 6: Outcome class distribution – Diabetes

Inference: Outcome is imbalanced (500 non-diabetic vs. 268 diabetic, roughly 65:35). This is mild but non-trivial imbalance, so precision, recall, and F1-score for the positive class should be reported alongside accuracy, and class weighting or resampling could improve recall on the minority (diabetic) class.

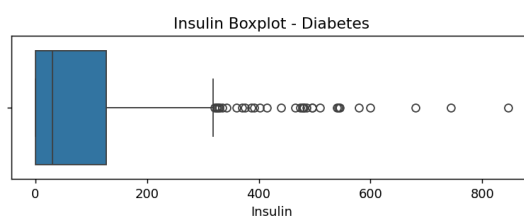


Figure 7: Insulin box plot – Diabetes

Inference: The IQR test flags 34 outliers in Insulin. Combined with the fact that 374 of 768 records (about 49%) have Insulin exactly equal to 0, this again points to placeholder zeros for unrecorded measurements rather than genuine physiological readings – this column will need explicit missing-value imputation rather than outlier removal.

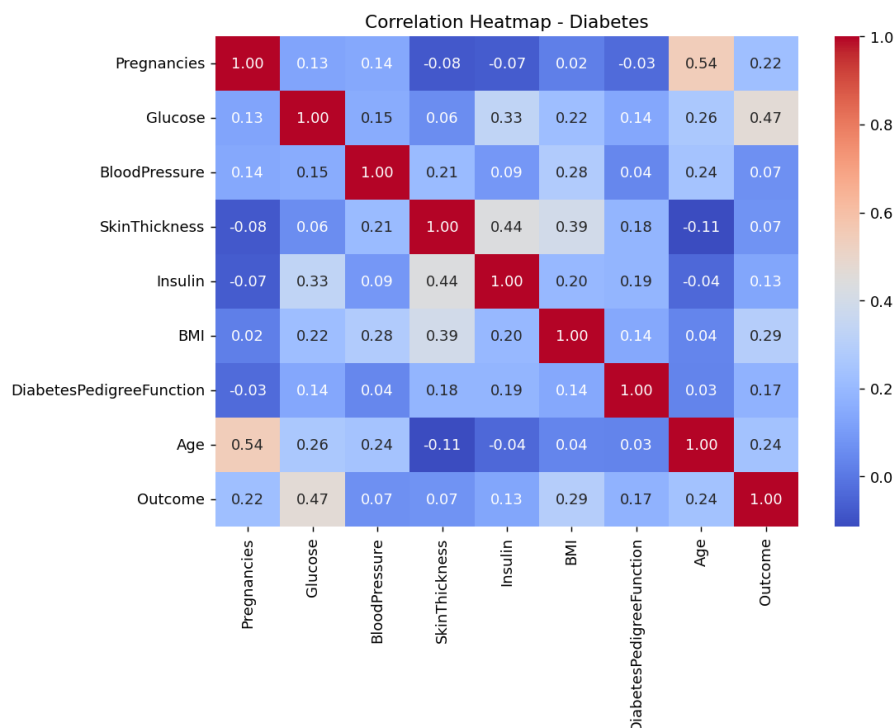


Figure 8: Correlation heatmap – Diabetes

Inference: Glucose shows the strongest correlation with Outcome (Pearson $r \approx 0.47$) of any single feature, consistent with its clinical role in diabetes diagnosis. Other features (BMI, Age, Pregnancies) show weaker but still positive associations, while no pair of predictors is correlated highly enough to suggest serious multicollinearity.

Loan Prediction Dataset

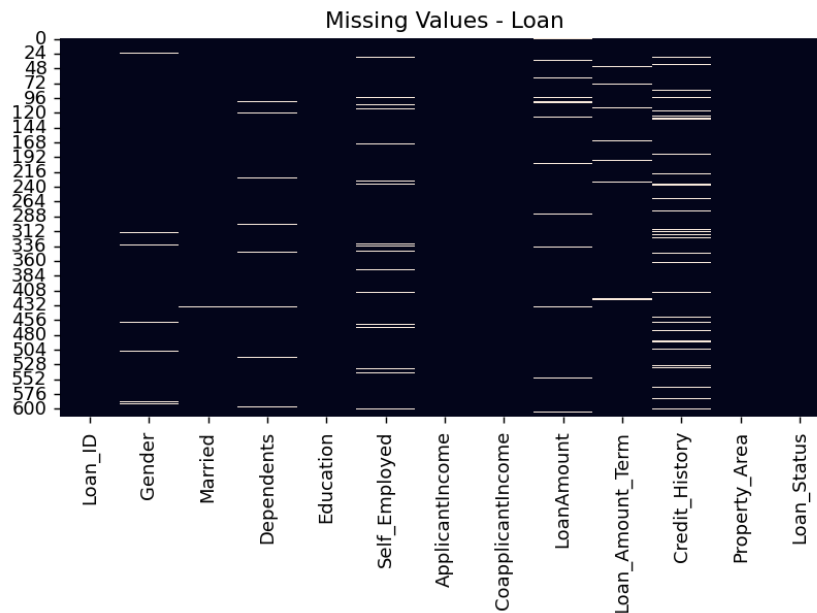


Figure 9: Missing value heatmap – Loan Prediction

Inference: Missing values are spread across seven columns, most severely Credit_History (50 missing), Self_Employed (32), and LoanAmount (22). Credit_History is typically the single strongest predictor of loan approval, so its missingness needs careful handling (mode/most-frequent imputation, or a separate “unknown” category) rather than row deletion, which would discard a meaningful fraction of the 614 records.

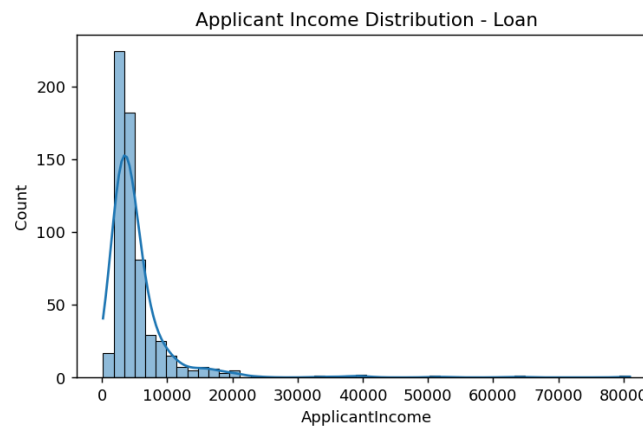


Figure 10: Applicant income distribution – Loan Prediction

Inference: Applicant income is heavily right-skewed (skewness ≈ 6.5), with most applicants clustered at lower incomes and a long tail of high earners. This strongly suggests a log transformation before feeding income into a regression model, and explains why 50 values are flagged as IQR outliers on the raw scale.



Figure 11: Loan status distribution – Loan Prediction

Inference: Loan_Status is imbalanced, with about 69% of applications approved (Y: 422) versus 31% rejected (N: 192). This imbalance should be reflected in the evaluation metric choice (F1/recall on the rejected class matter as much as overall accuracy) if the downstream task is framed as approval classification rather than amount regression.

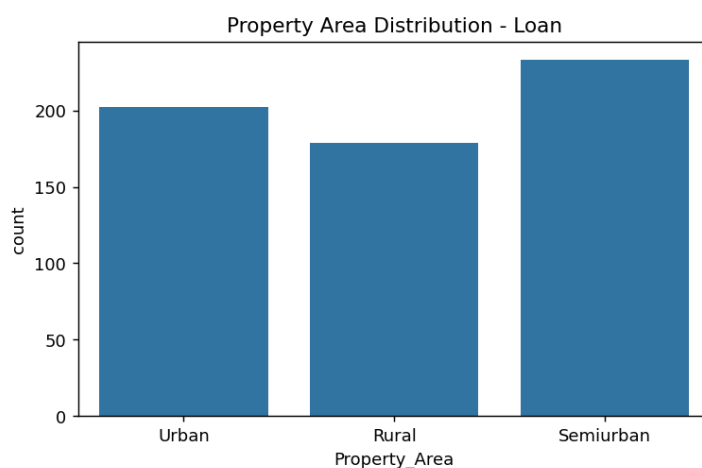


Figure 12: Property area distribution – Loan Prediction

Inference: Applications are reasonably distributed across the three property-area categories (Urban, Semiurban, Rural), with no single category dominating – so one-hot encoding this feature should not introduce severe sparsity or imbalance concerns.

Email Spam Dataset

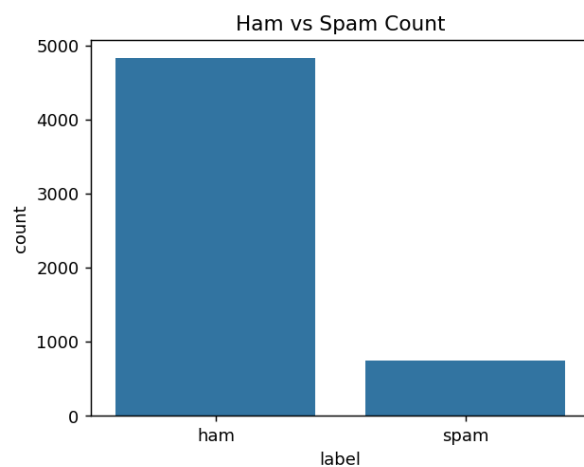


Figure 13: Ham vs. spam class distribution

Inference: The dataset is strongly imbalanced – 4,825 ham messages versus only 747 spam (about 87:13). A classifier trained naively on raw accuracy could reach $\sim 87\%$ by always predicting “ham,” so precision/recall/F1 on the spam class, or techniques such as class weighting, are essential for a meaningful evaluation.

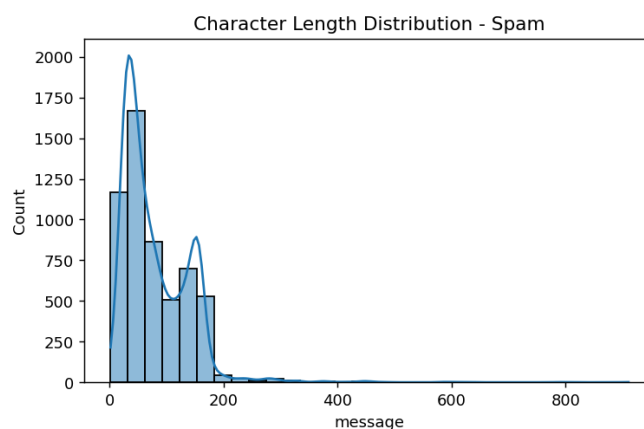


Figure 14: Character length distribution – Spam

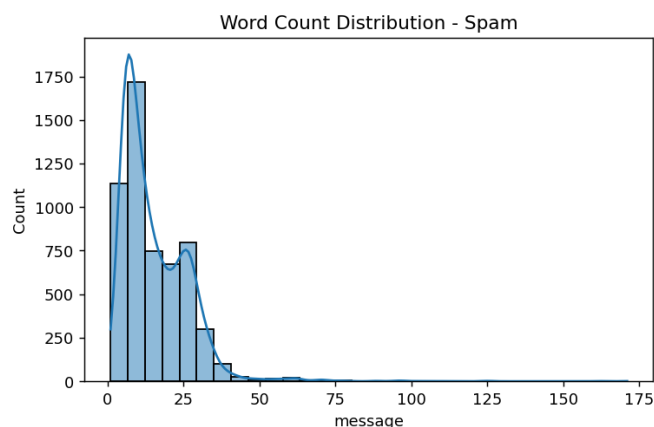


Figure 15: Word count distribution – Spam

Inference: Message length is itself discriminative: spam messages average about 139 characters versus roughly 71 for ham – almost double. Both distributions are right-skewed with a long tail toward longer messages, so character/word count could be added as an engineered feature alongside TF-IDF text features.

MNIST Dataset

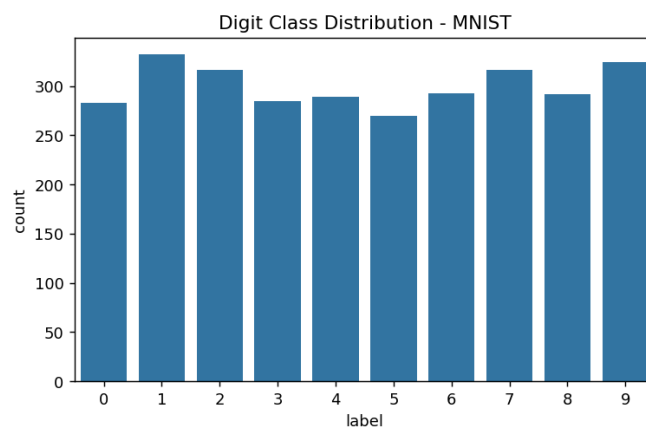


Figure 16: Digit class distribution – MNIST

Inference: The 10 digit classes are close to balanced in this sample, ranging from 270 (digit 5) to 332 (digit 1) examples – roughly a 20% spread. This near-balance means accuracy remains a reasonable primary metric, though a per-class confusion matrix is still useful given digits like 4/9 or 3/8 are visually similar and prone to confusion.

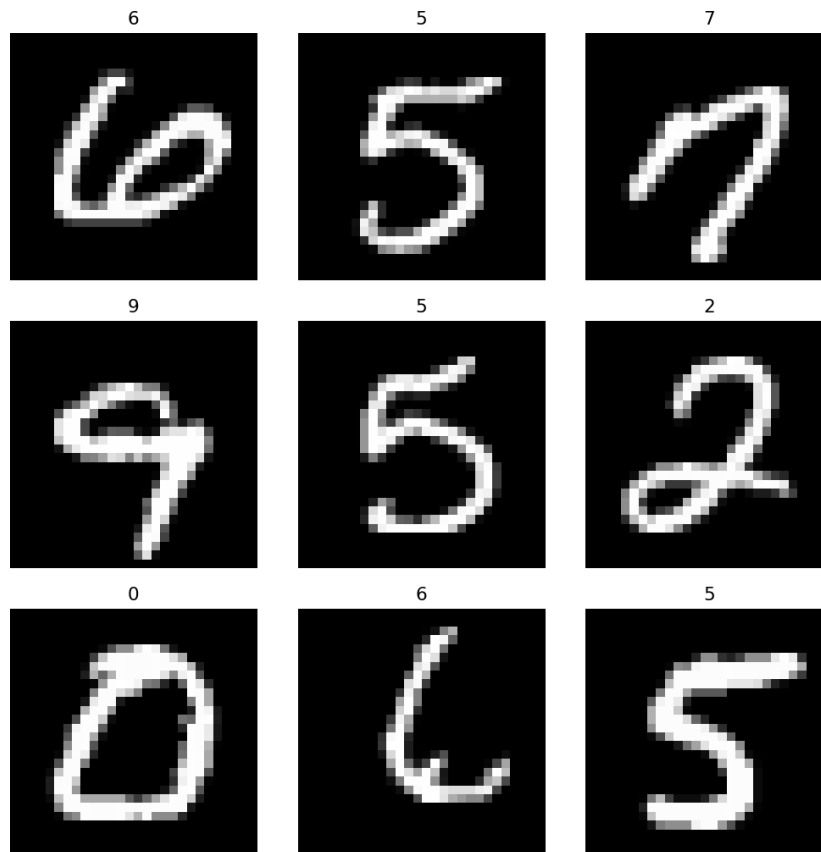


Figure 17: Sample digit images – MNIST

Inference: The sample grid confirms the images are correctly labelled, centered, grayscale 28×28 handwritten digits consistent with the standard MNIST format, with visible natural variation in stroke width and slant across samples of the same digit.

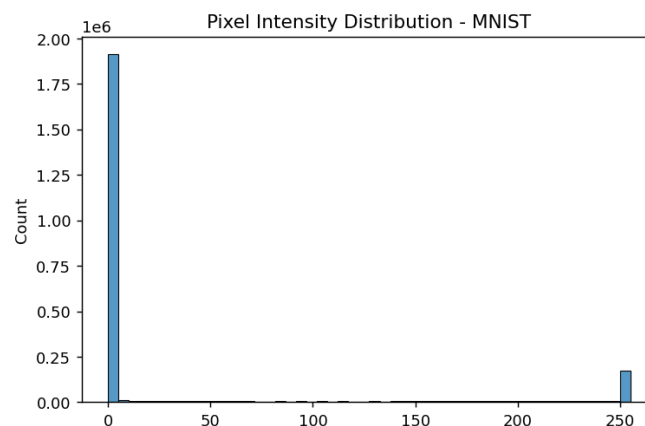


Figure 18: Pixel intensity distribution – MNIST

Inference: Pixel values range from 0 to 255 with a mean of about 33 and a large standard deviation (≈ 78), and only about 19% of pixels are non-zero. This confirms the images are mostly dark background with sparse bright strokes, and supports normalizing pixel values (dividing by 255) before feeding them into a CNN or any distance-based model.

5 Data Preprocessing

Explain:

- **Missing value handling:** Iris and MNIST require no imputation. Diabetes needs the disguised zero-placeholders in Glucose and Insulin replaced with median/mean imputation per Outcome class. Loan needs mode imputation for categorical columns (Gender, Self_Employed, Credit_History) and median imputation for LoanAmount.
- **Encoding:** Categorical features in Loan (Gender, Married, Education, Self_Employed, Property_Area) and the target labels in Diabetes/Spam/Iris/Loan require Label or One-Hot Encoding before modelling.
- **Feature scaling:** Diabetes and Loan numeric features (Glucose, BMI, ApplicantIncome) span very different ranges and should be standardized (**StandardScaler**) or normalized (**MinMaxScaler**). MNIST pixel values should be scaled to $[0, 1]$ by dividing by 255.
- **Train-test split:** Not performed in this experiment, which is limited to exploratory analysis; a typical 80:20 or 70:30 stratified split would be used in a subsequent modelling experiment.
- **Feature engineering (if applicable):** For Spam, message character length and word count (already explored in Section 4) could be added as engineered numeric features alongside text-vectorized (TF-IDF) features.

6 Machine Learning Algorithm

Based on the EDA findings, the following algorithms are recommended for each dataset (feature selection technique and full task identification is summarized in the table below).

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Iris	Supervised – Multi-class Classification	ANOVA F-test (SelectKBest with <code>f_classif</code>)	Logistic Regression / K-Nearest Neighbors / Decision Tree
Loan Amount Prediction	Supervised – Regression	Correlation analysis, ANOVA F-test (<code>f_regression</code>)	Linear Regression / Random Forest Regressor
Predicting Diabetes	Supervised – Binary Classification	Chi-square test (SelectKBest with <code>chi2</code>)	Logistic Regression / Random Forest Classifier
Classification of Email Spam	Supervised – Binary Classification (Text)	Chi-square test on TF-IDF/Bag-of-Words features	Naive Bayes / Support Vector Machine (SVM)

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Handwritten Character Recognition / MNIST	Supervised – Multi-class Classification (Image)	Variance thresholding / PCA (dimensionality reduction over pixel features)	Convolutional Neural Network (CNN) / SVM / Random Forest

Working principle (representative example – Logistic Regression): Models the probability of a binary/multi-class outcome using a linear combination of input features passed through a sigmoid (or softmax) function, trained by minimizing log-loss via gradient descent.

Mathematical formulation:

$$P(y = 1 | x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}$$

Advantages: Simple, interpretable, computationally cheap, works well when classes are near-linearly separable (as seen for setosa in the Iris pair plot).

Limitations: Struggles with non-linear decision boundaries and is sensitive to unscaled features and multicollinearity – both relevant given the skewed income feature in the Loan dataset and the correlated petal measurements in Iris.

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Selection

Not applicable for this experiment – Experiment 1 covers library exploration, dataset identification, and EDA only; no model was trained or tuned.

Parameter	Value
–	–

9 Results

Not applicable for this experiment – no model was trained. The equivalent deliverable for this experiment, the ML task/feature-selection/algorithm identification table, is provided in Section 6.

Metric	Value
Accuracy	–
Precision	–
Recall	–
F1-score	–
ROC-AUC	–

10 Result Visualization

Not applicable for this experiment – no model was trained, so no confusion matrix, ROC curve, or learning curve is available. The dataset-level visualizations (distributions, correlations, class balance) generated for this experiment are provided in Section 4.

11 Discussion

Across all five datasets, EDA surfaced issues that would materially affect modelling: disguised missing values (zeros) in Diabetes, genuine missing values concentrated in a few key columns in Loan, class imbalance in Diabetes/Loan/Spam, and a heavily right-skewed income feature in Loan. Iris and MNIST were the cleanest datasets, with no missing values and reasonably balanced classes. These findings directly justify the feature selection technique and algorithm choices recommended in Section 6 – e.g. chi-square/ANOVA tests for the tabular datasets, TF-IDF-based selection for text, and PCA/variance thresholding for the high-dimensional MNIST pixel features.

12 Conclusion

This experiment explored the core NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib/Seaborn functionality needed for a Machine Learning workflow, and used it to identify the ML task type, suitable feature selection technique, and suitable algorithm for five public datasets (Iris, Loan Amount Prediction, Diabetes Prediction, Email Spam Classification, and MNIST / Handwritten Character Recognition). Exploratory data analysis on the real datasets revealed concrete data-quality issues (disguised missing values, skewed distributions, class imbalance) that will directly inform preprocessing decisions in subsequent experiments.

13 References

- [1] R. A. Fisher, “The Use of Multiple Measurements in Taxonomic Problems,” *Annals of Eugenics*, 1936. Dataset accessed via https://scikit-learn.org/stable/datasets/toy_dataset.html
- [2] National Institute of Diabetes and Digestive and Kidney Diseases, “Pima Indians Diabetes Dataset,” UCI Machine Learning Repository, <https://archive.ics.uci.edu/>

- [3] Analytics Vidhya, “Loan Prediction Dataset,” Kaggle, <https://www.kaggle.com/datasets>
- [4] T. A. Almeida, J. M. G. Hidalgo, “SMS Spam Collection Dataset,” UCI Machine Learning Repository, <https://archive.ics.uci.edu/>
- [5] Y. LeCun, C. Cortes, C. J. C. Burges, “The MNIST Database of Handwritten Digits,” <http://yann.lecun.com/exdb/mnist/>
- [6] Scikit-learn Documentation, <https://scikit-learn.org/stable/>
- [7] Pandas Documentation, <https://pandas.pydata.org/docs/>